PHYS4055 Week 4 Tutorial Solutions

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Question 1: Entanglement distillation

Alice and Bob share n copies of the mixed state

$$\rho_{AB} = p\left|\phi^+\right\rangle\left\langle\phi^+\right| + (1-p)\left|\psi^+\right\rangle\left\langle\psi^+\right|, \qquad p > \frac{1}{2},$$

where

$$\left|\phi^+\right\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right), \qquad \left|\psi^+\right\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right)$$

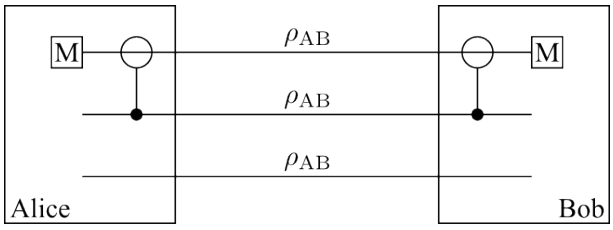
are Bell states. The goal of entanglement distillation is to use several imperfectly entangled pairs to obtain fewer pairs with higher entanglement.

For one step of the protocol, Alice and Bob take two copies of ρ_{AB}. On each side, they apply a CNOT gate, using the qubit from the first pair as the control qubit and the qubit from the second pair as the target qubit. The unitary operation of the CNOT gate can be written as

$$\hat{U}_{\text{CNOT}} = |0_C\rangle\langle 0_C| \otimes \hat{I}_T + |1_C\rangle\langle 1_C| \otimes \hat{\sigma}_{x,T},$$

that is, the target qubit is flipped if and only if the control qubit is $|1\rangle$.

After applying the two CNOT gates, Alice and Bob measure the target pair in the computational basis $\{|0\rangle, |1\rangle\}$ as shown in the figure below (where the first line is the target and the second line is the control). If they obtain the same measurement outcome, they keep the control pair; otherwise, they discard it.



(a) Calculate the state after one successful step:

  (a.1) Calculate the initial joint system.

  (a.2) Apply the CNOT gates.

  (a.3) Measure in the described basis and determine the resulting state.

(b) What is the probability that one distillation step is successful?

(c) Calculate the fidelity of the resulting state with respect to $|\phi^+\rangle$. Is it larger or smaller than the initial fidelity?

(d) Suppose that, after one successful step, Alice and Bob repeat the same procedure using pairs with the improved fidelity. What do you expect to happen to the fidelity after several successful rounds? What is the trade-off?